

PChome: Optimizing user experience and efficiency for ecommerce with Google Cloud

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About PChome

Founded in 1996, PChome is a leading ecommerce company in Taiwan with a wide range of services including web portal, logistics and payment. PChome 24h Shopping, its ecommerce platform, has around 13 million member users and offers 5 million+ products in various categories. In 2019, PChome launched a green shopping initiative to promote sustainability throughout its ecommerce supply chain.

Industries: Retail & Consumer Goods

development and operations efficie

Location: Taiwan

Google Cloud results

- Prevents service crashes caused by storage overload with Cloud Logging and Cloud Monitoring
- Reduces response time for DDoS attacks from three days to 10 minutes with Google Cloud Armor and Cloud Load Balancing
- Supports daily data analytics report generation for more effective marketing

Zero durir surg traffi by 2

(<https://cloud.google.com/logging/>),
Google Cloud (<https://cloud.google.com/>)
(<https://cloud.google.com/monitoring/>), Cloud
BigQuery (<https://cloud.google.com/bigquery/>)
Compute Engine
Logging (<https://cloud.google.com/logging/>)
Monitoring (<https://cloud.google.com/monitoring/>)
Kubernetes Engine
Cloud Storage (<https://cloud.google.com/storage/>)
Google Workspace
Compute Engine
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Cloud Firestore
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Kubernetes Engine
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Google Workspace
(<https://workspace.google.com/>)
Google Chat
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through

BigQuery

- Enables quick

internal

communication

for better

customer

services with

Chat

With more than 20 years of presence in ecommerce, PChome (<https://www.pchome.com.tw/>) is an industry leader in Taiwan. Founded in 1996, the company started as a web portal provider and launched its ecommerce business in 2000. PChome 24h Shopping (<https://24h.pchome.com.tw/>), its ecommerce platform, currently has around 13 million member users, which represents more than half of the population in the country.

The wide range of products offered on its platform embodies its commitment to providing an online marketplace where consumers can find everything they need in daily life. With an initial focus on consumer electronics products, PChome has greatly enriched its product variety over the past years to meet consumers' needs. PChome 24h Shopping now offers more than 5 million products in various categories, among which 2 million can be delivered within 24 hours in Taiwan.

"Our core value is 'users come first', and we've been improving our services on all fronts to bring the highest convenience to our customers," says Ascii

Huang, Technology and Product Director at PChome.

"With rich product selections, speedy delivery and personalized product recommendations, we want to make our platform the go-to place when users want to buy anything online."

PChome used to rent local data centers and deploy its services on on-premise servers. Due to complexities in the leasing model, it usually took the PChome team three to four months to acquire a new server, which limited its development speed and scalability. There was also a lack of good monitoring systems for servers and networks. This means that the PChome team could only rely on past experiences to predict whether certain hardware were crashing, which resulted in more challenging and less accurate infrastructure management.

To improve its service quality and operations, PChome decided to migrate a part of its services to a public cloud platform in late 2021. It eventually chose Google Cloud (<https://cloud.google.com/>) which can meet the country's information security requirements, especially as a Taipei Exchange-listed company.

"To ensure stable ecommerce services and facilitate our DevOps work, we needed to move our infrastructure from on-prem to cloud for greater scalability and better IT management," notes Huang.

"Google Cloud meets all our requirements for a public cloud platform and supports full security compliance with its data center in Taiwan."

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—Ascii Huang,
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Enhancing service stability and development efficiency with Compute Engine and GKE

During migration, the engineers of Google Cloud spent one to two hours every week to support the PChome IT team, offering them advice and guidance throughout the process. Within three months, PChome smoothly moved PChome 24h Shopping, its portal website, as well as its enterprise resource planning system, to Google Cloud. The company now taps on virtual machines (VMs) in Compute Engine (<https://cloud.google.com/compute>) to run services that contain constantly updated information like product stock, and microservices in Google Kubernetes Engine (<https://cloud.google.com/gke>) (GKE) for services that need to be frequently scaled up or down according to the traffic.

Huang notes that the quick launch of the instances in Compute Engine and GKE is particularly helpful for PChome. To prepare for sudden traffic surges during shopping events like Singles' Day, which can bring in 15-20 times more users than usual, the team requires a large number of instances ready in a short period of time. With Compute Engine and GKE, the PChome team is able to launch 8,000 computing instances in 15 minutes, which allows the company

to test its ecommerce website thoroughly and smoothly tackle spikes in traffic when they occur.

"We've tried different offerings and found that Compute Engine and GKE support the shortest computing instance launching time. This way, we can simulate traffic surges as realistically as possible and prepare ourselves for different scenarios," he adds. "We used to encounter sporadic service crashes during shopping festivals, but since the migration to Google Cloud, our ecommerce platform has never experienced any downtime due to high traffic."

The swift access to more computing resources in Compute Engine and GKE has also helped PChome enhance its development efficiency. Before, its development team needed to wait for at least three months for a new server to arrive. Now, the team can quickly launch new VMs and microservices whenever there is a need.

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Director, PChome*

Supporting effective data-driven decision making with BigQuery

PChome previously relied on a self-built system to process and analyze its data. However, its data processing speed and analytics capabilities were

not enough for the company to implement data-driven strategies effectively. For example, it used to take the system four to five hours to generate sales reports for one retailer brand, which made it impossible for PChome to provide useful sales data to its retailer partners on PChome 24h Shopping.

By adopting BigQuery.

(<https://cloud.google.com/bigquery>) to analyze data and generate insights, PChome is now able to have the analysis of the previous day's data ready every day before its employees begin work. With more timely and detailed data analytics, the company's marketing team can produce more effective marketing strategies. One example is that it started sending coupons to dormant customers who are identified through data analytics, and the customer retention rate has been greatly improved. PChome can now also provide daily sales and advertising performance reports to its retail partners, so that they can improve their sales numbers and advertising strategies.

"The sublime data analytics capabilities of BigQuery have enabled us and our retailer partners to implement more data-driven strategies and reach better business results," notes Huang. "Without BigQuery, we would have needed several months to change the structure of our previous data analytics system to support the similar analysis speed and range."



The PChome 24h shopping platform has 2 million products in stock that can be delivered in Taiwan within 24 hours.

Increasing visibility and facilitating collaboration for better customer services

The monitoring and security tools of Google Cloud have also enabled PChome to improve its service stability. The company leverages Firestore (<https://cloud.google.com/firestore>) and Cloud Storage (<https://cloud.google.com/storage>) to store its website data while using Cloud Monitoring (<https://cloud.google.com/monitoring>) and Cloud Logging (<https://cloud.google.com/logging>) to monitor the performance of its cloud infrastructure. When its data were stored in on-premise hardware, due to low visibility, it was challenging for the team to accurately detect the hard drives that were becoming full. Overloaded hard drives would then lead to service crashes a few times every year. Since Cloud Monitoring and Cloud Logging provide real-time performance data and can send alerts when storage reaches certain thresholds, the PChome

team can easily prevent incidents caused by data overload.

PChome has also started leveraging the Google Workspace (<https://workspace.google.com/business/>) suite to support more efficient internal collaboration. Its employees now use Docs (<https://workspace.google.com/products/docs>) and Sheets (<https://workspace.google.com/products/sheets>) to create files for internal use. On top of being easy to share as attachments through Gmail (<https://workspace.google.com/products/gmail>) and Meet (<https://workspace.google.com/products/meet/>), the access authorization can be effortlessly managed with the Google Workspace accounts.

The company has also set up different channels in Chat (<https://workspace.google.com/products/chat/>) to enable faster communication across departments. For example, PChome engineers have established a system for scraping data from popular online forums to uncover negative comments by its customers. When a complaint is found, the monitoring systems would immediately send it to the relevant channel for product managers or the customer service team to address. By responding to users' complaints quickly, PChome has seen a noticeable improvement in customer satisfaction.

Leveraging AI to optimize ecommerce customer experiences

Moving forward, PChome plans to use the retail solutions (<https://cloud.google.com/solutions/retail>) of Google Cloud, powered by artificial intelligence (AI),

to increase the automation of its operations and offer more innovative features. Its DevOps team is now using Vision AI (<https://cloud.google.com/vision>) to build a new feature that enables image search on its ecommerce website, which will be available in late 2023. Huang says that expanding the use of cloud tools is also in line with the company's sustainability strategy and helps reduce its carbon footprint.

"Google Cloud has helped us smoothly reach our digital transformation goals by enhancing our team's agility and work efficiency to deliver better ecommerce user experiences," he notes. "We're now equipped with excellent tools to continue improving our services and fulfilling our users' expectations for ecommerce innovation."

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